ALTRUS – ADAPTIVE LIFE-SUPPORT AND THERAPEUTIC ROBOTIC UNIT SYSTEM

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BSc (Hons) degree in Information Technology Specializing in Software Engineering

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DECLARATION

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Date ……..

ABSTRACT

Emotionally aware human-robot interaction demands systems capable of perceiving, interpreting, and responding to human affective states in a manner that is contextually sensitive, temporally coherent, and conversationally meaningful. Existing approaches are constrained by three structural limitations: (i) individual sensing modalities are processed in isolation without adaptive integration, (ii) conversational and emotional context across interaction turns is disregarded, and (iii) emotion recognition terminates at classification without generating emotionally conditioned responses. Together, these gaps prevent the realization of fluid, intelligent interaction in assistive robotic applications.

This dissertation presents RoboEmotion, a unified, modular, and real-time framework that simultaneously addresses all three limitations within a single privacy-preserving architecture. The system integrates CNN-based facial emotion recognition, LSTM-based speech emotion recognition using MFCC features, an adaptive confidence-weighted multimodal fusion layer with dynamic modality reweighting, a context-aware conversation manager that maintains a rolling emotional and dialogue history, and a locally deployed Large Language Model for emotion-conditioned response generation that eliminating cloud dependency and ensuring full data privacy compliance.

Experimental evaluation demonstrates that multimodal fusion consistently outperforms unimodal baselines and maintains robust performance under degraded environmental conditions. Context-aware response generation produced significant improvements in conversation coherence, response appropriateness, and overall user satisfaction compared to context-free baselines. End-to-end latency on consumer-grade hardware confirms real-time deployment viability within natural conversational turn-taking intervals.

The framework contributes a deployable, extensible foundation for emotionally intelligent assistive robotics, with practical relevance to eldercare, mental health support, and social companion applications.

Keywords: Multimodal Emotion Detection, Human-Robot Interaction, Affective Computing, Facial Expression Recognition, Speech Emotion Recognition, Large Language Models, Context-Aware Response Generation, Assistive Robotics

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List of abbreviations

Abbreviation	Full Form
API	Application Programming Interface
ASR	Automatic Speech Recognition
CNN	Convolutional Neural Network
CPU	Central Processing Unit
CREMA-D	Crowd-Sourced Emotional Multimodal Actors Dataset
FER	Facial Emotion Recognition
FACS	Facial Action Coding System
FPS	Frames Per Second
GDPR	General Data Protection Regulation
GPU	Graphics Processing Unit
HRI	Human-Robot Interaction
HIPAA	Health Insurance Portability and Accountability Act
HCI	Human-Computer Interaction
IEEE	Institute of Electrical and Electronics Engineers
LLM	Large Language Model
LSTM	Long Short-Term Memory
MFCC	Mel-Frequency Cepstral Coefficient
MOS	Mean Opinion Score
NFR	Non-Functional Requirement
RAM	Random Access Memory
RAVDESS	Ryerson Audio-Visual Database of Emotional Speech and Song
ReLU	Rectified Linear Unit
RMS	Root Mean Square
ROS	Robot Operating System
SDK	Software Development Kit
SER	Speech Emotion Recognition
SNR	Signal-to-Noise Ratio
TRL	Technology Readiness Level

Abbreviation	Full Form
TTS	Text-to-Speech
UAT	User Acceptance Testing
VAD	Voice Activity Detection
VRAM	Video Random Access Memory

1. INTRODUCTION

The relationship between humans and machines has undergone a profound transformation over the past two decades. Where once robots were confined to factory floors, executing repetitive physical tasks in isolation from human presence, they are now being deployed in hospitals, schools, rehabilitation centers, and private homes. That environments defined not by mechanical precision but by human complexity, vulnerability, and emotional need. This shift in deployment context demands a corresponding shift in capability: the machines entering these spaces must not merely perform tasks but must engage with people, and to engage meaningfully with people is, inevitably, to engage with emotion.

Emotion is not peripheral to human communication. It is central to it. It shapes how people speak, what they mean, what they need, and how they respond. A caregiver who cannot read the distress beneath a patient's calm words, or who fails to notice the anxiety behind a child's silence, is not an effective caregiver. The same principle applies to robotic systems deployed in these contexts. A robot companion that responds to a grieving elderly patient with the same communicative register it uses when that patient is cheerful has failed at the most fundamental level of social intelligence that not technically, but humanly. The consequences of such failure in therapeutic and care settings are not merely awkward; they can be actively harmful, eroding user trust, reducing engagement, and undermining the therapeutic value of the interaction itself.

Recognizing this, affective computing that the field concerned with enabling machines to perceive, interpret, and respond to human emotional states that has emerged as one of the most consequential research frontiers in human–robot interaction. Substantial progress has been made in individual recognition domains: deep convolutional neural networks can classify facial expressions with near-human accuracy, and recurrent neural architectures trained on speech corpora can identify emotional content from vocal features alone. Yet despite these advances, a critical gap persists between what emotion recognition systems can detect and what emotionally intelligent robotic systems must do.

The gap is threefold. First, most deployed systems are unimodal. They process either visual or acoustic signals, but not both in an integrated, adaptive manner. Human emotional expression, however, is inherently multimodal: facial cues, vocal prosody, gesture, and linguistic content are not independent channels but complementary facets of a single communicative act. A system that processes only one channel is not merely incomplete. It is brittle, failing precisely when conditions degrade and robust perception is most needed. Second, existing systems treat each

conversational turn as an isolated event, classifying the emotion expressed in the current utterance without reference to what preceded it. Yet emotion in conversation is not a snapshot, it is a trajectory. A user's emotional state at any given moment is shaped by the history of the exchange, and a response that ignores that history, however accurate in its per-turn classification, will feel disconnected and socially unintelligent. Third, and most fundamentally, current systems terminate at classification. They produce an emotion label and stop that leaving unaddressed the question of what the system should actually say or do in response to what it has detected. The translation of emotional perception into emotionally appropriate action remains, in most existing frameworks, an unsolved problem.

The emergence of Large Language Models systems such as Gemma, Llama 3, and Mistral 7B — presents a transformative opportunity to address this third gap. These models possess the contextual reasoning, tonal flexibility, and generative fluency necessary to produce responses that are not merely grammatically coherent but emotionally resonant. When grounded in structured, real-time emotional context, they offer a viable path toward robotic systems that do not merely detect emotion but respond to it with genuine sensitivity. The challenge lies in integrating these capabilities. Multimodal emotion detection, contextual history tracking, and LLM-driven response generation into a unified, real-time, privacy-preserving framework deployable on consumer-grade robotic hardware without dependency on external cloud services.

This dissertation presents RoboEmotion, a framework designed to meet precisely that challenge. By unifying real-time multimodal emotion detection, adaptive fusion, conversational context management, and locally deployed language model inference within a single modular architecture, RoboEmotion represents a substantive step toward robotic systems capable of emotionally intelligent interaction.

1.1 Background Literature

The RoboEmotion framework draws on five interconnected research disciplines. This section reviews theoretical foundations and empirical developments in each area to establish the context for the research gap analysis.

1.1.1 Facial Emotion Recognition

The scientific study of facial emotion expression predates the field of affective computing by more than a century, tracing its origins to Darwin's observations on the continuity of emotional expression across species. It was, however, the psychologist Paul Ekman whose empirical cross-cultural studies in the 1960s and 1970s established the framework that continues to define automated facial emotion recognition to this day. Through a series of studies conducted across culturally isolated populations that including pre-literate communities in Papua New Guinea with no prior exposure to Western media. Ekman demonstrated that the recognition of six discrete emotional expressions was consistent across cultures, leading to the proposition that happiness, sadness, anger, fear, disgust, and surprise constitute a set of biologically universal, pancultural emotional signals [10]. These findings were formalized in the Facial Action Coding System (FACS), a comprehensive anatomical taxonomy encoding facial muscle movements as discrete Facial Action Units (AUs), which provided automated FER research with both a theoretical foundation and a practical labelling vocabulary that remains standard across the literature three decades later.

The translation of this psychological framework into computational systems underwent a decisive acceleration with the advent of deep learning. Prior to the deep learning era, FER systems relied on handcrafted feature representations. Local Binary Patterns, Histogram of Oriented Gradients, Gabor filter responses that combined with classical classifiers such as Support Vector Machines. While these approaches demonstrated reasonable performance on constrained laboratory datasets, they exhibited significant fragility under real-world variation in illumination, head pose, and facial occlusion. The introduction of deep Convolutional Neural Networks transformed this landscape. CNNs learn hierarchical feature representations directly from pixel data, progressively extracting increasingly abstract facial structures from edges and textures at early layers to expression-specific activation patterns at deeper layers without requiring expert-designed feature engineering.

The FER2013 dataset, introduced by [7] at the 2013 ICML Challenges in Representation Learning competition, played a pivotal role in establishing CNN-based FER as the field's

dominant paradigm. Comprising 35,887 grayscale facial images at 48×48-pixel resolution across seven emotion categories, the dataset enabled systematic large-scale benchmarking of deep architectures and drove a sustained period of rapid performance improvement. [1] subsequently demonstrated that lightweight CNN architectures that specifically designed to balance representational capacity against computational cost that can achieve real-time FER inference at over 30 frames per second on GPU hardware while maintaining competitive classification accuracy. This contribution is of direct significance to the present work: robotic deployment contexts impose strict latency constraints that heavier architectures cannot satisfy, and the existence of lightweight CNN designs that meet both accuracy and speed requirements is a necessary condition for the real-time FER module developed here.

A notable limitation of the original FER2013 benchmark is its susceptibility to annotation noise, arising from the crowd-sourced labelling process in which annotators frequently disagreed on ambiguous expressions particularly those belonging to the fear, disgust, and contempt categories, whose facial action unit configurations partially overlap. The FER2013 Plus variant addresses this limitation through systematic crowd-sourced re-annotation, assigning soft probability labels across emotion categories rather than hard single-class assignments, thereby producing a more reliable training signal and reducing the risk of a model learning to fit mislabeled examples. The present work employs FER2013 Plus for precisely this reason.

Architectural innovation beyond the standard CNN has continued to advance FER performance. Jiang et al. [2] demonstrated that Residual Network (ResNet) architectures, which introduce shortcut connections allowing gradients to bypass convolutional blocks during training, achieve improved generalization under conditions of partial facial occlusion, variable head pose, and uncontrolled illumination that conditions directly characteristic of real-world HRI environments where users move freely, may be partially obscured, and interact under variable lighting. More recently, attention-based Vision Transformer (ViT) architectures have achieved state-of-the-art results on multiple FER benchmarks by modelling long-range dependencies between facial regions through self-attention mechanisms, capturing holistic expression configurations that local convolutional filters may miss. However, the substantially higher computational and memory requirements of transformer architectures that typically an order of magnitude larger than lightweight CNNs in parameter count. Currently preclude their deployment in real-time embedded robotic contexts without aggressive quantization or knowledge distillation, making them unsuitable for the present application without significant engineering overhead.

A deeper limitation of the FER literature that is rarely foregrounded but is of considerable practical significance is the temporal shallowness of most deployed systems. Frame-level classification is the dominant paradigm that processes each video frame independently, producing an emotion label for a static snapshot of facial configuration without modelling the temporal dynamics of expression: the way an expression builds from neutrality, reaches its apex, and decays back over the duration of an utterance. Human observers do not perceive emotion from static frames; they perceive it from the unfolding of expression over time, and the onset and offset trajectories of an expression carry significant information about its authenticity, intensity, and communicative intent. Dynamic FER approaches that including 3D CNNs that treat video clips as spatiotemporal volumes, and ConvLSTM architectures that couple convolutional feature extraction with recurrent temporal modelling that address this limitation at the cost of substantially increased computational complexity. In the present work, frame-level CNN classification is adopted to satisfy the real-time processing constraint, with temporal coherence maintained at the system level through confidence-weighted averaging across consecutive frames at the fusion layer that a pragmatic design decision that acknowledges the trade-off between temporal richness and deployment feasibility.

1.1.2 Speech Emotion Recognition

While the face is perhaps the most immediately legible channel of emotional expression, the voice carries emotional information of equal richness and, in many contexts, greater reliability. Vocal emotion is encoded across multiple acoustic dimensions simultaneously: the fundamental frequency of the voice that perceived as pitch that rises with excitement and falls with sadness; speech rate accelerates under stress and slows under grief; vocal energy and intensity vary with arousal; and the spectral distribution of energy across frequency bands shifts in characteristic ways across emotional states. Unlike facial expression, which requires visual line of sight and is subject to deliberate social masking, vocal emotional cues are more difficult to suppress and remain informative even when a speaker is making conscious efforts to appear calm. Speech emotion recognition (SER) is the computational discipline concerned with extracting and classifying these cues automatically, independent of the linguistic content of the utterance a distinction that is both technically significant and practically important, since the same words can carry entirely different emotional meaning depending on how they are spoken.

The central challenge in SER is the choice of acoustic feature representation: the transformation of a raw audio waveform into a compact numerical descriptor that preserves emotionally

discriminative information while discarding irrelevant variation. The most widely adopted representation in both classical and deep learning SER pipelines is the Mel-Frequency Cepstral Coefficient (MFCC). MFCCs are computed by applying a bank of triangular filters spaced on the Mel frequency scale a perceptually motivated transformation that compresses higher frequencies in accordance with the non-linear frequency resolution of the human auditory system to the short-time Fourier spectrum of the speech signal, then applying a logarithmic energy transformation and a Discrete Cosine Transform to produce a compact cepstral representation. The resulting feature vectors capture the spectral envelope of the speech signal in a form that is both computationally tractable and perceptually meaningful, encoding the timbral and prosodic characteristics most strongly associated with emotional variation. Across decades of SER benchmarking, MFCC features have consistently demonstrated strong discriminability across emotion categories, and they constitute the primary acoustic representation employed in the present work.

Prior to the deep learning era, SER systems combined handcrafted acoustic features — including MFCCs, pitch contours, energy envelopes, and zero-crossing rates with classical machine learning classifiers such as Support Vector Machines, Gaussian Mixture Models, and Hidden Markov Models. These approaches achieved reasonable performance on controlled laboratory datasets but exhibited the same fragility under naturalistic conditions that characterized pre-deep learning FER systems: their handcrafted features encoded assumptions about which acoustic properties were emotionally relevant, assumptions that proved increasingly inadequate as evaluation moved from acted, studio-recorded datasets toward spontaneous, conversational speech. The shift to deep learning, which replaces expert-designed feature engineering with learned representations, produced substantial and consistent performance improvements across SER benchmarks.

[3] provided an influential demonstration of this shift, showing that end-to-end deep learning architectures combining convolutional feature extraction from raw or minimally processed audio with Long Short-Term Memory (LSTM) temporal modelling achieve state-of-the-art SER performance on the IEMOCAP and RAVDESS benchmarks [15], surpassing the best handcrafted feature pipelines by substantial margins. The LSTM is particularly well suited to SER because emotion in speech is expressed over time a single acoustic frame carries little information in isolation, but the trajectory of pitch, energy, and spectral features across the duration of an utterance constitutes a rich emotional signature. Standard LSTMs process this trajectory in a single temporal direction; the bidirectional LSTM extension, which processes the

input sequence simultaneously in forward and backward directions and concatenates the resulting hidden state sequences, captures the full prosodic arc of an utterance that including offset patterns and the relationship between utterance-final features and overall emotional trajectory and has been consistently shown to improve SER accuracy relative to unidirectional processing. The bidirectional LSTM architecture forms the basis of the speech emotion recognition module developed in this work.

An important theoretical extension to utterance-level SER is the incorporation of conversational context. [4] demonstrated that models incorporating the emotional labels and acoustic features of preceding utterances in a multi-turn dialogue achieve significantly higher classification accuracy than models processing each utterance in isolation, with the improvement most pronounced for subtle emotional transitions where the current utterance alone is acoustically ambiguous. This finding has direct implications for the design of the present system: it establishes that emotional classification is not merely a single-utterance problem but a sequential inference problem in which the history of the interaction constitutes evidence for the interpretation of the current state. This theoretical grounding motivates the emotional trajectory tracking component of the context-aware conversation management module developed in RoboEmotion, which maintains a rolling history of detected emotion labels across interaction turns and provides this history as structured context to the response generation layer.

Perhaps the most practically significant and insufficiently acknowledged limitation of the SER literature is the cultural and linguistic specificity of its benchmark datasets. The three most widely used SER evaluation corpora — RAVDESS [15], CREMA-D, and IEMOCAP [14] share a critical characteristic: they were recorded exclusively with North American English speakers, predominantly from Western, educated, industrialized, rich, and democratic backgrounds. Prosodic conventions that the patterns of pitch variation, speech rate, energy modulation, and pause placement that convey emotional meaning are not universal. They are culturally shaped, language-specific, and subject to substantial regional variation even within a single language. A model trained to recognize the prosodic signature of sadness as expressed by a North American English speaker has learned a culturally specific emotional code, not a universal one. When applied to speakers from different linguistic communities including Sri Lankan English speakers, as evaluated in the present work such models encounter prosodic, phonological, and intonational patterns systematically absent from their training distribution, producing predictable and potentially severe performance degradation. This cross-cultural generalization failure is not a minor technical limitation but a fundamental equity challenge for the global deployment of SER-

dependent systems: a system that works well for one demographic and poorly for another reproduces and technologically embeds the biases of its training data. The present work confronts this challenge directly through cross-cultural evaluation and targeted adaptation, and identifies the collection of regionally representative speech emotion corpora as the most critical open problem for equitable SER deployment.

1.1.3 Multimodal Emotion Fusion

Human beings do not experience or express emotion through a single channel. When a person is distressed, their face contracts, their voice tightens, their posture collapses, and the words they choose reflect their inner state simultaneously, redundantly, and in ways that are partially but not entirely correlated. This redundancy is not accidental; it is evolutionarily functional, ensuring that emotional signals remain legible even when individual channels are degraded by noise, distance, or social suppression. A computational system that processes only one of these channels is therefore not merely incomplete in a technical sense that it is operating against the grain of how emotional communication is structured. Multimodal emotion recognition is the field concerned with building systems that honor this structure, integrating information across sensing modalities to produce emotional estimates that are more accurate, more robust, and more ecologically valid than any single-channel approach can achieve.

The theoretical justification for multimodal integration rests on two distinct but complementary principles. The first is information redundancy: different modalities carry partially overlapping emotional signals, meaning that when one channel produces an uncertain or erroneous classification, another may carry sufficient information to correct it. The second is information complementarity: beyond their overlap, different modalities also encode emotionally relevant information that is genuinely independent aspects of emotional state that are visible in the face but not the voice, or audible in speech but not legible from facial configuration alone. Together, these principles predict that the performance ceiling of any multimodal system will exceed that of its best unimodal component a prediction that has been confirmed so consistently across the empirical literature that it now functions less as a hypothesis to be tested and more as a design principle to be implemented [5].

The central engineering question in multimodal fusion is not whether to combine modalities but how and at what level of abstraction. Three broad strategies have been established in the literature. Feature-level fusion, also called early fusion, concatenates the raw or processed feature representations from multiple modalities into a single combined vector before any

classification is performed. This approach allows the model to learn cross-modal correlations at the feature level and can capture subtle interaction effects between modalities that decision-level approaches cannot access. Its disadvantage is sensitivity to modality absence: if one sensing channel fails or is unavailable, the entire feature vector is corrupted, and the model trained on complete multimodal inputs may perform poorly or fail entirely. Decision-level fusion, also called late fusion, takes the opposite approach: each modality is processed independently through its own classification pipeline, and the resulting predictions are combined through a fixed or adaptive rule after classification. This architecture is more robust to modality absence when one channel fails, its contribution can simply be zeroed or down-weighted without disrupting the other and is more computationally flexible, since modality-specific models can be developed, trained, and updated independently. Hybrid fusion integrates both levels, combining feature-level cross-modal interaction with decision-level combination, and can in principle achieve the advantages of both at the cost of increased architectural complexity.

The choice between these strategies is not purely theoretical; it is strongly conditioned by deployment constraints. [5] conducted a systematic empirical evaluation of late fusion strategies for audio-visual emotion recognition, examining fixed-weight averaging, majority voting, and adaptive confidence-weighted fusion across a range of environmental conditions. Their central finding was that adaptive late fusion in which the contribution weight of each modality is dynamically adjusted based on the confidence score of its per-sample prediction, rather than fixed globally consistently outperforms both fixed-weight and voting approaches, particularly under conditions where individual modalities are degraded. The intuition is straightforward: when the facial recognition channel is operating in poor lighting and producing low-confidence predictions, those predictions should contribute less to the final decision than the voice channel, which may be unaffected by lighting. A fixed-weight scheme cannot respond to this situational asymmetry; an adaptive scheme does so automatically. This finding directly motivates the confidence-weighted adaptive fusion strategy adopted in RoboEmotion, where modality weights are computed dynamically at each inference step based on the output confidence of the facial and speech recognition modules.

More recent work has explored the application of transformer-based cross-modal attention architectures to multimodal emotion fusion. [6] demonstrated that models employing cross-modal attention mechanisms in which the query, key, and value representations used in self-attention are drawn from different modalities, enabling each modality's representation to be directly conditioned on the other to achieve state-of-the-art performance on the CMU-MOSI and

IEMOCAP benchmarks by capturing rich inter-modal dependencies that concatenation-based and decision-level approaches cannot model. These results represent the current frontier of multimodal emotion recognition performance. However, the inference latency and GPU memory requirements of transformer-based fusion architectures at this scale are incompatible with the real-time, consumer-hardware deployment constraints of the present work. The adaptive late fusion strategy adopted here represents a deliberate and principled trade-off: it sacrifices the cross-modal representational richness of attention-based approaches in exchange for the computational efficiency, architectural modularity, and graceful degradation under modality failure that real-world robotic deployment requires.

1.1.4 Conversational AI and Large Language Models

For most of the history of natural language processing, building systems capable of coherent, contextually sensitive dialogue required elaborate hand-engineered pipelines: rule-based dialogue managers, slot-filling architectures, retrieval-based response selectors, and domain-specific knowledge bases that had to be constructed and maintained by human experts. These systems were brittle by design, operating reliably only within the narrow domains for which they had been explicitly programmed and failing unpredictably when users deviated from expected interaction patterns. The aspiration of genuinely open-domain, contextually aware conversation remained largely unrealized.

The introduction of the Transformer architecture by [11] in 2017 marked the beginning of a fundamental change in this landscape. The Transformer replaced the sequential recurrent processing of earlier neural language models with a fully parallel self-attention mechanism, in which every token in an input sequence attends simultaneously to every other token, allowing the model to capture long-range dependencies without the vanishing gradient problems that limited recurrent architectures. This parallelism made it possible to train on corpora orders of magnitude larger than had previously been feasible, and the models that emerged from training at this scale exhibited capabilities that had not been explicitly designed or anticipated: coherent multi-turn dialogue, instruction following, contextual reasoning across extended passages, stylistic flexibility, and the ability to generalize to tasks never encountered during training. These emergent capabilities, now associated with the class of systems known as Large Language Models, transformed the practical horizon of conversational AI.

For the purposes of emotionally adaptive response generation in human-robot interaction, LLMs offer three capabilities of particular relevance. The first is prompt-conditioned tonal adaptation:

when provided with structured information about a user's detected emotional state in the system prompt, an LLM can modulate its response register, word choice, and communicative stance accordingly, producing outputs that are empathetic under sadness, encouraging under anxiety, and celebratory under joy, without requiring any task-specific fine-tuning of the model weights. The second is multi-turn contextual coherence: by processing the full conversation history within their context window, LLMs maintain thematic and logical consistency across extended exchanges in a way that single-turn response generators cannot, naturally producing responses that acknowledge what has been said before rather than treating each turn as an isolated input-output pair. The third is local deploy ability: open-weight models including Llama 3 [12], Gemma [13], and Mistral 7B can be served entirely on consumer hardware through inference frameworks such as Ollama, enabling privacy-preserving on-device inference without dependency on external cloud APIs, subscription costs, or network connectivity. This last capability is not merely a convenience but a prerequisite for deployment in healthcare and eldercare contexts, where the transmission of patient interaction data to external servers raises serious regulatory and ethical concerns under frameworks such as HIPAA and GDPR.

Despite these substantial capabilities, a critical gap persists between what existing LLM-based conversational systems do and what emotionally grounded human-robot interaction requires. Commercially deployed conversational agents, and the majority of affective dialogue systems described in the research literature, derive their emotional context from text alone: they analyze the sentiment or emotional content of the user's typed or transcribed words and use this analysis to condition their responses. This approach carries a fundamental limitation that is rarely foregrounded but is of considerable practical significance. Language is only one of several channels through which emotional state is communicated, and it is often the least reliable. People routinely use words that misrepresent or actively conceal their emotional state: a grieving patient who says they are fine, an anxious child who insists everything is normal, an elderly user who minimizes distress to avoid burdening their caregiver. In each of these cases, the true emotional signal is carried not by the words but by the face and the voice, and a system that attends only to linguistic content will be systematically deceived by the very users who most need emotionally sensitive support. Grounding LLM-based response generation in real-time audio-visual emotion detection rather than text sentiment alone is therefore not an incremental improvement but a qualitative shift in the nature of the system's emotional awareness. Achieving this integration within a real-time, locally deployable, privacy-preserving framework constitutes the central technical contribution of the present work.

1.1.5 Affective Computing in Assistive Robotics

The application of affective computing to assistive and social robotics has been a subject of active research since Breazeal's pioneering work on the Kismet robot in the late 1990s, which demonstrated that robots capable of expressing and responding to human emotions elicit more natural and engaging interactions from human partners. Commercial social robots including SoftBank Robotics' Pepper and NAO platforms, Furhat Robotics' conversational robot head, and healthcare-oriented platforms such as the PARO therapeutic robot have since demonstrated the potential for emotionally responsive robotic interaction to improve patient engagement, mood, and therapeutic outcomes in eldercare and rehabilitation settings [8].

A systematic review by [8] established the fundamental evaluation framework for affective interactive systems, identifying real-time processing capability, robustness under naturalistic conditions, and perceived emotional appropriateness of responses as the three primary design criteria for practical deployment. Critically, Cowie et al. demonstrated that perceived emotional appropriateness of responses that are not raw emotion classification accuracy is the dominant determinant of user acceptance and interaction effectiveness. This finding has profound implications for system design prioritization: engineering resources invested in improving response appropriateness (through context tracking and LLM integration) may yield greater practical benefit than equivalent resources invested in marginal accuracy improvements to emotion classifiers.

[9] documented a systematic ecological validity gap: FER and SER models trained on acted, laboratory-collected datasets achieve significantly lower accuracy in naturalistic, spontaneous interaction settings, where expressions are subtler, more ambiguous, and more variable. This gap represents a fundamental challenge for the deployment of laboratory-developed emotion recognition systems in real-world assistive robotics applications, and motivates the robustness evaluation under naturalistic degraded conditions conducted in the present work.

1.2 Research Gap

Systematic analysis of the five literature areas reviewed in Section 1.1 reveals four persistent and interconnected limitations in existing emotion-aware interactive systems that collectively define the research gap addressed by RoboEmotion:

Gap 1 - Unimodal Isolation

The majority of deployed emotion-aware interactive systems process visual or audio inputs independently, without a unified multimodal decision-making framework that adaptively integrates information from both channels. This architectural isolation produces brittle systems: FER systems fail under occlusion and adverse lighting; SER systems degrade under background noise and speaker accent variation. No existing production-ready framework implements adaptive multimodal fusion with dynamic modality reweighting that compensates for channel-specific degradation in real time.

Gap 2 - Context Blindness

Existing emotion-aware systems treat each conversational turn as an independent event, processing the current utterance or video frame without reference to the conversational history or the emotional trajectory of preceding turns. This produces responses that may be locally appropriate to the instantaneous detected emotion but are globally incoherent with the user's evolving emotional state across the interaction — a critical failure mode in therapeutic and care contexts where emotional continuity and contextual sensitivity are clinically essential [4].

Gap 3 - Classification Without Response

Current emotion recognition systems terminate at classification, producing emotion labels without generating corresponding emotionally adaptive conversational outputs. Despite the clear technical and practical complementarity between multimodal emotion detection and LLM-based response generation, no existing framework integrates these two capabilities in a unified, real-time, locally deployable pipeline. The result is a fundamental gap between emotion perception and emotionally intelligent action [3][6].

Gap 4 — Privacy and Deployment Constraints

Most multimodal emotion recognition research is evaluated offline on curated laboratory datasets. Systems that integrate cloud-based LLM APIs introduce network latency, per-query

costs, and data privacy risks that are fundamentally incompatible with healthcare data protection requirements (HIPAA, GDPR). No existing framework simultaneously achieves real-time multimodal performance, on-device privacy-preserving LLM inference, and modular extensibility for diverse robotic hardware configurations.

Table 1 presents a structured comparison of representative prior systems against the proposed RoboEmotion framework across all four gap dimensions.

System / Study	Multimodal Fusion	Conv. Context	LLM Response Gen.	Real-Time	On-Device Privacy
FER CNN	No	No	No	Yes	Yes
SER LSTM	No	No	No	No	Yes
AV Fusion	Yes (Late)	No	No	No	Yes
Transformer	Yes (Attention)	Partial	No	No	Yes
GPT-4 / Cloud LLM Chatbots	No	Yes (Text only)	Yes (Cloud)	Yes	No
SoftBank Pepper SDK	Partial	Limited	Limited	Yes	Partial
RoboEmotion (Proposed)	Yes (Adaptive)	Yes (A/V + Text)	Yes (Local LLM)	Yes	Yes

Table 1 : Comparison of Related Systems Against Proposed Framework

1.3 Research Problem

The core research problem addressed by this work is formally stated as follows:

> *"Current emotion-aware systems for human–robot interaction lack a unified framework that simultaneously performs real-time multimodal emotion detection from facial and vocal cues, maintains conversational and emotional context across interaction turns, and generates emotion-conditioned natural language responses through locally deployable large language models — thus failing to support fluid, emotionally intelligent, and privacy-preserving robotic interaction in assistive care applications."*

This gap is particularly consequential in assistive robotics contexts including dementia care, autism support, post-traumatic stress rehabilitation, and social skill development where emotional responsiveness, interaction continuity, and conversational coherence are not merely desirable but are clinically essential components of effective therapeutic engagement. The failure to close the emotion perception-to-response loop represents a barrier not only to technical quality but to the social utility and therapeutic effectiveness of emotion-aware robotic systems.

1.4 Research Objectives

This research aims to design, implement, and evaluate RoboEmotion: a modular, real-time framework for emotion-aware HRI addressing all four identified research gap dimensions. The specific objectives are:

1. Develop a CNN-based facial emotion recognition module classifying six emotion categories from real-time webcam input, achieving validation accuracy exceeding 75% on the FER2013 Plus benchmark through principled architecture design, data augmentation, and regularization.
2. Develop an LSTM-based speech emotion recognition module utilizing 40 MFCC features from RAVDESS and CREMA-D datasets, with systematic cross-cultural evaluation and targeted binary classification adaptation for reliable Sri Lankan deployment.
3. Design an adaptive multimodal fusion layer combining facial and vocal predictions through confidence-weighted averaging with dynamic modality reweighting under channel unavailability or degradation.
4. Implement a context-aware conversation management system maintaining a rolling 10-turn window of conversation history and emotional trajectory, formatted as structured LLM-ready prompts.
5. Integrate locally deployed large language models (Gemma, Llama, Mistral via Ollama) for emotion-conditioned response generation, with full on-device inference and configurable plugin-based output delivery.
6. Evaluate the framework comprehensively: emotion recognition performance metrics; user acceptance testing (N=30) comparing context-aware vs. context-free baselines; system latency, resource utilization, and robustness across four environmental conditions.

2. METHODOLOGY

Building a system that operates reliably in real time, integrates multiple independently complex components, and must ultimately be evaluated against both technical performance metrics and human experience requires more than sound engineering at the component level. It requires a development process that is structured enough to manage complexity and risk, yet flexible enough to accommodate the iterative refinement that emerges when theoretical design meets practical implementation. This chapter presents the complete methodology underlying the RoboEmotion framework, covering the development approach, system architecture, individual component design and implementation, commercialization analysis, and testing strategy.

The development followed an iterative Agile methodology organized into four sequential phases. The first phase, Requirements Analysis, established the functional and non-functional specifications of the system through review of the HRI literature, structured consultation with domain professionals, and analysis of the constraints imposed by real-world robotic deployment contexts. The second phase, Component Implementation and Unit Testing, involved the independent design, implementation, training, and validation of each system module against its individual acceptance criteria before integration. The third phase, System Integration, assembled the independently validated components into a unified, concurrently executing pipeline and verified correct end-to-end behavior through structured integration testing. The fourth phase, Evaluation, assessed the complete framework against the research objectives through quantitative performance measurement, robustness testing across controlled environmental conditions, and a structured user acceptance study. Each phase produced defined deliverables that provided the inputs and constraints for the subsequent phase, enabling systematic risk management and early detection of design issues before they propagated across the system.

2.1 System Architecture

The RoboEmotion system is structured as a five-stage layered processing pipeline: Input → Detection → Fusion → Generation → Output. Each layer is designed as an independently testable, modular software component. The pipeline runs in a multithreaded real-time execution environment in which video processing, audio processing, and LLM inference threads operate concurrently and communicate through thread-safe Python Queue objects.

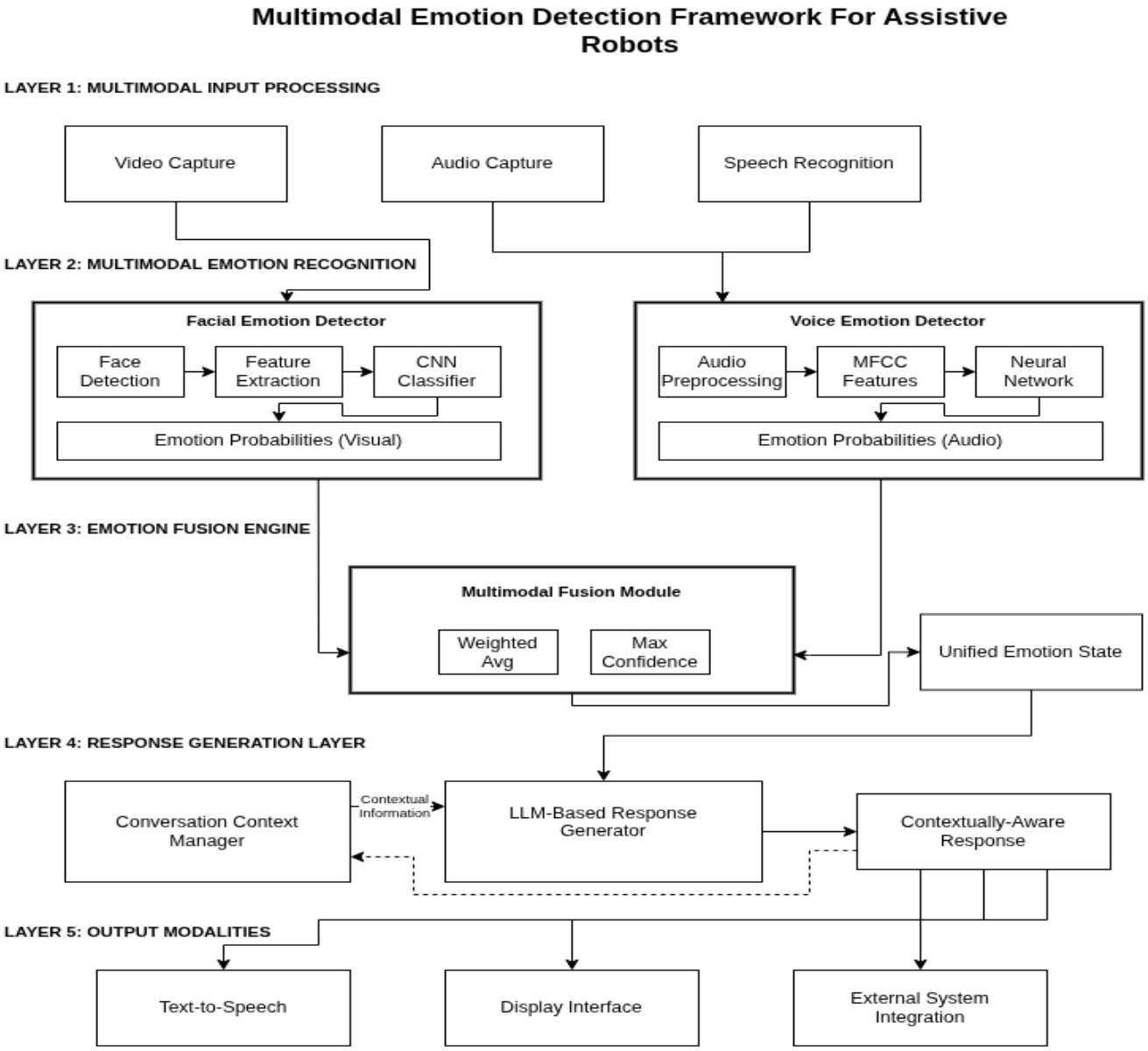

Figure 1 : System Diagram For Emotion Detection Framework

2.2 Software Solution

The RoboEmotion framework is implemented as a self-contained, modular software subsystem within the broader ALTRUS platform, responsible for all emotion perception, contextual reasoning, and affective response generation capabilities of the assistive robot. While the overall ALTRUS system provides the robotics middleware, navigation, and hardware abstraction layers through ROS2 Humble, the RoboEmotion framework operates as a set of independently executing processing modules that interface with the platform through standardized plugin callbacks and communication interfaces. This design ensures that the emotion intelligence layer remains loosely coupled from the underlying robotic infrastructure, enabling deployment across different hardware configurations without architectural modification.

The software solution is structured around a five-stage layered processing pipeline. That are input, detection, fusion, generation, and output in which each stage is implemented as a discrete, independently testable module communicating with adjacent stages through thread-safe queue interfaces. The pipeline executes in a multithreaded real-time environment in which video processing, audio processing, and language model inference threads operate concurrently, preventing the computational cost of any single stage from blocking the real-time processing of others.

The input layer manages two parallel data acquisition pipelines. The visual pipeline captures video through a standard USB webcam at 640x480 pixel resolution and 30 frames per second using OpenCV, applying Haar Cascade face detection and preprocessing each detected face region through grayscale conversion and pixel normalization prior to emotion classification. The audio pipeline captures speech signals at 16 kHz in mono format using the sound device library, employing RMS-based Voice Activity Detection to segment individual utterances and routing each detected speech segment concurrently to both the Whisper automatic speech recognition pipeline for transcription and the MFCC feature extraction pipeline for emotion analysis.

The emotion detection layer processes visual and audio inputs through two independently trained deep learning models. Facial emotion recognition is performed by a custom Convolutional Neural Network trained on the FER2013 Plus dataset, classifying 48x48 pixel grayscale face images into six emotion categories and producing a per-class probability distribution with an associated confidence score. Speech emotion recognition is performed by a bidirectional Long Short-Term Memory network trained on the RAVDESS and CREMA-D datasets, operating on

40 MFCC features extracted per audio segment and producing emotion probability distributions for each detected utterance.

The multimodal fusion layer integrates the outputs of both detection modules into a unified emotional estimate through confidence-weighted adaptive averaging, dynamically reweighting modality contributions based on per-prediction confidence scores and automatically falling back to single-modality inference when either channel is unavailable or degraded. This adaptive strategy ensures that the system continues to produce reliable emotional estimates across the range of environmental conditions encountered in real-world deployment, including poor lighting, background noise, and periods of user silence.

The context-aware conversation management module maintains two persistent data structures across the duration of each interaction session: a rolling deque of the last ten conversational turns storing both utterance transcripts and system responses, and an emotional trajectory history recording the detected emotion label and fusion confidence at each turn. These histories are serialized at each response generation step into a structured prompt that provides the language model with both the conversational and emotional context necessary to generate responses that are coherent across turns rather than reactive to individual utterances in isolation.

Response generation is performed by large language models deployed entirely on-device through the Ollama inference framework, with Gemma 3:1b as the primary configuration and support for Llama 3 and Mistral 7B as alternative backends. The on-device deployment eliminates dependency on external cloud services, ensuring that all user interaction data remains private and that the system remains operational without network connectivity. Generated responses are delivered through text-to-speech synthesis for audio output, and through a plugin callback interface that enables integration with the robotic hardware output systems provided by the ALTRUS platform, including LED control, gesture systems, and animation controllers.

The complete solution is implemented in Python 3.10, using TensorFlow and Keras for model training and inference, OpenCV for visual processing, Librosa for audio feature extraction, and the Python threading library for concurrent pipeline execution.

2.2.1 Requirement Analysis

The requirements for the RoboEmotion component were derived through a systematic analysis of the functional demands of emotionally aware human-robot interaction, the technical constraints of real-time robotic deployment, and the integration requirements imposed by the broader ALTRUS platform architecture. This process identified four principal challenges that the component must address:

- **Subsystem Integration Complexity:** Facial emotion recognition, speech emotion recognition, automatic speech recognition, multimodal fusion, and language model inference each carry distinct computational profiles, data formats, and timing requirements. Coordinating these subsystems within a single real-time pipeline without introducing blocking dependencies or data synchronization failures requires deliberate architectural design.
- **Absence of Reusable Affective HRI Frameworks:** Existing research prototypes in the emotion recognition literature are typically designed for offline evaluation on curated datasets rather than for deployment in live, conversational robotic contexts, leaving a significant gap between available tooling and the requirements of the present work.
- **Coupling of Perception and Response Generation:** Emotion classification and language model inference operate on fundamentally different timescales. Ensuring that the emotional state detected in one interaction turn is correctly propagated to and reflected in the corresponding generated response requires careful management of shared state across concurrent processing threads.
- **Privacy-Preserving On-Device Operation:** Healthcare and eldercare deployment contexts impose strict constraints on the transmission of user interaction data to external services, requiring that all inference, including large language model inference, be performed locally on consumer-grade hardware without cloud dependency.

To address these challenges, the RoboEmotion framework is designed around four guiding principles:

- **Modular Layered Architecture:** Each processing stage is independently testable and replaceable, enabling isolated development, unit testing, and component-level validation prior to system integration.
- **Configuration-Driven Behavior:** Fusion strategies, context window sizes, language model backends, and output delivery mechanisms are adjustable through configuration

parameters without requiring code modification, supporting rapid adaptation across deployment contexts.

- **Multithreaded Execution Model:** Video processing, audio processing, and language model inference execute in dedicated concurrent threads connected through thread-safe queue interfaces, decoupling the timing constraints of individual stages and ensuring that no single component blocks the real-time operation of others.
- **Plugin-Based Output Interface:** A standardized callback interface enables integration with diverse robotic hardware platforms without coupling the emotion intelligence layer to any specific hardware configuration, consistent with the modular interoperability conventions of the broader ALTRUS platform.

2.2.2 Feasibility Study

A feasibility analysis was conducted across three dimensions to assess the viability of the RoboEmotion framework prior to implementation.

- **Technical Feasibility:**

  The technical implementation of RoboEmotion is grounded in widely adopted, actively maintained open-source technologies. Deep learning frameworks including TensorFlow and Keras provide robust support for CNN and LSTM model development, training, and real-time inference. OpenCV and Librosa are mature, well-documented libraries with established track records in computer vision and audio processing applications respectively. The Whisper automatic speech recognition model and the Ollama local inference framework are both actively developed and provide reliable, production-grade performance on consumer hardware. The multithreaded Python execution model, combined with thread-safe queue-based inter-module communication, is a well-understood concurrency pattern that has been validated in real-time processing applications. Collectively, these technologies provide a technically sound foundation for the real-time, multimodal, privacy-preserving pipeline that RoboEmotion requires.

- **Operational Feasibility:**

  The RoboEmotion component is designed to operate without specialist hardware or infrastructure beyond a standard webcam, microphone, and consumer-grade

computing device with a GPU. The system requires no network connectivity during operation, no cloud service subscriptions, and no proprietary software licenses. The plugin-based output interface and configuration-driven behavior model ensure that the component can be integrated into diverse robotic platforms and adapted to different deployment contexts without requiring modifications to the core codebase. End users interact with the system through natural speech and facial expression, requiring no training or technical knowledge, while developers can extend the system through well-defined interfaces without requiring deep familiarity with the underlying emotion recognition models.

- **Economic Feasibility:**

  The RoboEmotion framework relies exclusively on open-source software components, eliminating licensing costs across the full technology stack. The hardware requirements are met by consumer-grade devices, with the system validated on a configuration costing significantly less than specialized robotic sensing platforms. The locally deployed language model inference eliminates ongoing per-query API costs associated with cloud-based language model services, making the operational cost of running the system effectively zero beyond the initial hardware investment. The modular architecture further reduces long-term maintenance costs by enabling individual components to be updated or replaced independently as better models or tools become available, without requiring full system redevelopment.

2.3 Initial Data Collection

Initial data collection for the RoboEmotion component focused on two primary domains relevant to multimodal emotion recognition.

- **Facial Expression Data**

  Visual data for facial emotion recognition was sourced from the FER2013 Plus dataset, a large-scale benchmark comprising grayscale facial images annotated across six discrete emotion categories: happiness, sadness, anger, fear, disgust, and surprise. The dataset provides 28,709 training images, 3,589 validation images, and 3,589 test images, offering sufficient volume and class diversity for training a robust convolutional neural network classifier. Images were captured under varied illumination conditions, head poses, and demographic backgrounds, providing the distributional diversity necessary to support generalization beyond controlled laboratory conditions. The FER2013 Plus variant was selected over the original FER2013 dataset specifically because its crowd-sourced re-annotation process reduces label noise, producing a more reliable training signal particularly for emotion categories whose facial action unit configurations are visually similar and therefore prone to annotator disagreement.

- **Audio and Speech Data**

  Audio data for speech emotion recognition was sourced from two established benchmark corpora. The RAVDESS dataset comprises 1,440 recordings from 24 professional actors across eight emotion categories, recorded under controlled studio conditions at consistent audio quality. The CREMA-D dataset provides 7,442 audio clips from 91 demographically diverse actors across six emotion categories, offering broader speaker coverage and greater variability in recording conditions. From each audio recording, 40 Mel-Frequency Cepstral Coefficient features were extracted per segment using the Librosa audio processing library, producing the temporal feature matrices used as input to the LSTM-based speech emotion classifier. Additionally, real-time audio signals captured through a standard microphone during system operation provided the live inference data stream processed by the speech recognition pipeline during deployment.

Both datasets were processed through their respective preprocessing pipelines prior to model training, with normalization, augmentation, and temporal standardization applied to ensure consistent input representations across the full range of recording conditions present in the training corpora.

2.4 Proposed System

The ALTRUS platform is a unified assistive robotics system that integrates four specialized subsystems into a cohesive, interoperable architecture designed to support intelligent, human-facing robotic deployment across healthcare, eldercare, and rehabilitation environments.

The Smart Navigation Framework provides adaptive indoor mobility through simultaneous localization and mapping, enabling the robot to build and continuously update representations of its operating environment while executing dynamic path planning and obstacle avoidance in real time.

The IoT Firmware Generator automates the creation, compilation, and deployment of embedded firmware for ESP32-based devices, supporting modular sensor integration for real-time health monitoring applications including heart rate tracking and activity recognition.

The RoboEmotion Framework provides the platform's emotion-aware interaction capabilities, integrating multimodal emotion detection from facial and vocal cues, contextual conversation management, and locally deployed large language model inference to enable responses that are sensitive to the emotional state of the user across the full duration of an interaction session.

The Supervisory Middleware coordinates communication and data exchange between all subsystems, providing fault tolerance, system state monitoring, and secure local logging to ensure reliable operation across extended deployment periods.

Together, these subsystems deliver a platform with three core interaction capabilities. First, multimodal user interaction through voice, facial expression, gesture, and graphical interface inputs, ensuring accessibility across diverse user populations and interaction preferences. Second, real-time health monitoring through wearable ESP32-based sensor networks, enabling continuous tracking of physiological indicators relevant to assistive care. Third, emotion-aware conversational response generation that adapts to the detected affective state of the user, supporting therapeutic and social engagement objectives in care settings. The platform is further designed for fault-tolerant operation, maintaining core functionality and safe behavior in the

event of individual subsystem failures, ensuring reliability in the uncontrolled conditions characteristic of real-world assistive robotics deployment.

2.5 Project Requirements

2.5.1 Functional Requirements

The functional requirements specific to the RoboEmotion framework define the core behaviors that the emotion-aware interaction subsystem must provide within the broader ALTRUS platform.

- **FR1: Multimodal Emotion Detection:** The system must detect human emotional states in real time by processing facial expression data captured through a webcam and vocal tone features extracted from microphone audio, combining both modalities through an adaptive fusion strategy to produce a unified emotional estimate with an associated confidence score.
- **FR2: Emotion-Aware Response Generation:** Based on the detected emotional state and the accumulated conversational and emotional history of the current interaction session, the system must generate natural language responses whose tone, register, and content are conditioned on the user's affective state, delivered through text-to-speech synthesis and configurable robotic hardware output callbacks.

These two requirements constitute the primary functional scope of the RoboEmotion framework. Their realization depends on four supporting behaviors that are implicit within FR1 and FR2 but warrant explicit specification for the purposes of component-level verification.

- The system must perform face detection and localization from live video input and classify detected faces into six discrete emotion categories with per-class confidence scores.
- The system must detect speech segments from continuous audio input through voice activity detection, extract 40 MFCC features per segment, and classify the emotional content of each detected utterance.
- The system must maintain a rolling history of the last ten conversational turns, recording both utterance transcripts and detected emotion labels at each turn, and serialize this history into structured prompts for language model inference.
- The system must perform all inference, including large language model response generation, entirely on-device without transmitting any user interaction data to external systems or cloud services.

2.5.2 Non-functional Requirements

Non-functional requirements define system quality attributes and operational constraints.

- **NFR1: Reliability** – The system must operate continuously with minimal failures and provide stable performance under varying conditions.
- **NFR2: Fault Tolerance** – The system must detect, isolate, and recover from faults without causing complete system failure.
- **NFR3: Performance** – System response time for navigation, emotion detection, and command execution must remain within real-time constraints (low latency).
- **NFR4: Scalability** – The architecture must support the addition of new sensors, modules, and features without major redesign.
- **NFR5: Modularity** – System components must be loosely coupled and independently deployable to support maintainability and reuse.
- **NFR6: Security** – Data transmission and system logs must be protected against unauthorized access and tampering.
- **NFR7: Usability** – The system must provide developer-friendly tools and interfaces to simplify configuration and deployment.
- **NFR8: Maintainability** – The system must support easy updates, debugging, and extension through clean architecture and documentation.

2.5.3 User Requirements

User requirements describe the expectations and needs of the individuals who interact directly with the RoboEmotion framework, spanning both end-users in assistive care contexts and the developers who build, extend, and deploy the system.

- **UR1: Natural Interaction:** Users must be able to interact with the RoboEmotion system through natural spoken conversation and facial expression without requiring any specialized commands, scripted phrases, or technical knowledge. The system must respond to spontaneous, unstructured speech in the same way it responds to deliberate interaction.

- **UR2: Emotionally Appropriate Responses:** The system must provide responses that feel emotionally relevant and contextually sensitive to the user's current affective state. Users must not be required to explicitly state their emotion; the system must infer it automatically from facial and vocal cues and reflect it in the tone and content of its responses.
- **UR3: Real-Time Conversational Feedback:** The system must deliver responses within a timeframe that feels natural and conversational to the user, without noticeable delays that disrupt the flow of interaction or reduce the perceived attentiveness of the robot.
- **UR4: Conversational Continuity:** The system must demonstrate awareness of what has been said and felt across the duration of the interaction session, producing responses that acknowledge the emotional trajectory of the conversation rather than treating each utterance as an isolated event. Users must feel that the system is listening and remembering, not merely reacting.
- **UR5: Minimal Technical Knowledge Required:** The system must be fully usable by non-technical users, including elderly individuals and patients in rehabilitation settings, without requiring any configuration, setup, or understanding of the underlying technology. All interaction must occur through natural communication modalities that require no prior training.
- **UR6: Privacy Assurance:** Users must be able to trust that their facial expressions, voice, and conversational content are processed locally on the device and are not transmitted to external services. The system must provide this assurance by design rather than by policy, ensuring privacy through its architecture rather than relying on user-facing terms and conditions.
- **UR7: Developer-Friendly Framework:** The system must provide a clean, well-documented, and extensible software framework that enables developers to integrate new emotion recognition models, swap language model backends, add output delivery plugins, and adapt the pipeline to new robotic platforms with minimal effort. Configuration-driven behavior, standardized interfaces, and modular component boundaries must reduce integration complexity and lower the barrier to contribution and customization across diverse development contexts.

2.5.4 System Requirements

System requirements define the hardware and infrastructure necessary to support the operation of the RoboEmotion framework within the broader ALTRUS platform.

- **SR1: Computing Unit:** The system must include a computing platform capable of running concurrent deep learning inference, large language model generation, and real-time audio-visual processing simultaneously. A consumer-grade device with a dedicated GPU is the recommended configuration, with CPU-only operation supported at reduced latency performance.
- **SR2: Vision System:** A camera module must be present and operational, capable of capturing video at a minimum resolution of 640x480 pixels and 30 frames per second, to support continuous real-time facial emotion detection. The camera must provide sufficient image quality under the ambient lighting conditions of the intended deployment environment.
- **SR3: Audio System:** A microphone module must support audio capture at a minimum sampling rate of 16 kHz in mono format for speech emotion recognition and automatic speech transcription. A speaker module or text-to-speech output device must be available for delivery of generated verbal responses to the user.
- **SR4: Storage:** Sufficient local storage must be available to accommodate deep learning model weights for the facial CNN and speech LSTM classifiers, the locally deployed large language model files, and session interaction logs generated during operation. A minimum of 10 GB of available storage is required for the primary Gemma 3:1b model configuration.
- **SR5: Memory:** A minimum of 8 GB of system RAM is required for concurrent execution of the full processing pipeline, with 16 GB recommended for stable operation of the language model inference layer alongside the emotion detection and fusion modules.
- **SR6: Operating System:** The system must run on Ubuntu 20.04 or later, or Windows 10 or later, with Python 3.10 or higher installed and GPU driver support configured for TensorFlow and Ollama inference where a discrete GPU is available.
- **SR7: Network Independence:** The system must be capable of full operation without any active network connection. All model inference, including large language model generation, must execute locally, ensuring that the system remains fully functional in network-restricted environments such as hospital wards and care facilities.

2.5.5 Software Requirements

Software requirements define the tools, frameworks, and libraries necessary to implement and operate the RoboEmotion framework within the ALTRUS platform environment.

- **SWR1: Core Programming Language:** Python 3.10 or later must be used as the primary implementation language for all RoboEmotion pipeline components, including model inference, audio-visual processing, fusion logic, context management, and language model integration, providing the flexibility and library ecosystem required for rapid development and extensibility.
- **SWR2: Deep Learning Framework:** TensorFlow 2.15 or later with Keras must be used for the implementation, training, and real-time inference of the CNN-based facial emotion recognition model and the LSTM-based speech emotion recognition model, providing GPU-accelerated inference support and model serialization capabilities.
- **SWR3: Computer Vision Library:** OpenCV 4.8 or later must be used for webcam video capture, Haar Cascade face detection, face region extraction, and image preprocessing operations including grayscale conversion, resizing, and pixel normalization.
- **SWR4: Audio Processing Library:** Librosa must be used for MFCC feature extraction from speech segments, providing the mel-scale filterbank computation, logarithmic energy transformation, and discrete cosine transform operations required to produce the acoustic feature representations used by the speech emotion classifier.
- **SWR5: Audio Capture Library:** The sounddevice library must be used for real-time microphone audio capture at 16 kHz in mono format, with RMS-based Voice Activity Detection implemented over sliding audio frames for speech segment detection and utterance boundary identification.
- **SWR6: Automatic Speech Recognition:** The OpenAI Whisper base model must be used for speech-to-text transcription of detected utterances, providing the conversational transcript content stored in the context management module for language model prompt construction.
- **SWR7: Local LLM Inference Framework:** Ollama must be used for on-device large language model serving, supporting Gemma 3:1b as the primary configuration with Llama 3 and Mistral 7B as alternative backends, providing a unified REST API for emotion-conditioned response generation without external network dependency.

- **SWR8: Robotics Middleware Integration:** ROS2 Humble must be supported as the communication and integration layer through which the RoboEmotion framework interfaces with other ALTRUS subsystems, including the navigation framework and supervisory middleware, via standardized topic and service interfaces.
- **SWR9: Concurrency Framework:** The Python threading standard library must be used to implement the multithreaded pipeline execution model, with thread-safe Queue objects managing data exchange between the video processing, audio processing, fusion, and language model inference threads.
- **SWR10: DevOps and Deployment Tools:** Docker and Docker Compose must be used to containerize the RoboEmotion runtime environment, ensuring consistent dependency resolution and execution behavior across development, testing, and production deployment stages. Git must be used for version control of all source code and configuration assets.

2.5.6 Personnel Requirements

Personnel requirements define the expertise needed for development and maintenance.

- **PR1: Robotics Developers** – Responsible for implementing ROS2 nodes, navigation, and system integration.
- **PR2: Embedded Systems Engineers** – Responsible for ESP32 firmware development and IoT device integration.
- **PR3: AI/ML Engineers** – Responsible for developing and optimizing machine learning models for perception and analysis.
- **PR4: UI/UX Developers** – Responsible for designing user interfaces for interaction and monitoring.
- **PR5: System Testers** – Responsible for validating system performance, reliability, and usability through testing.

2.6 Commercialization

The commercialization strategy of the ALTRUS framework is designed to maximize both **technical adoption and market scalability** by targeting multiple customer segments through a **dual-market approach**. This strategy ensures that the project is not limited to a single product offering but instead evolves into a **sustainable ecosystem of software, hardware, and services**.

1) B2B Software Platform

The primary commercialization pathway is the **ALTRUS Framework Suite as a software platform**, targeting:

- Robotics startups
- Assistive robot manufacturers
- Research and development laboratories
- Universities and academic institutions
- IoT developers and system integrators

These users often face challenges related to **integration complexity, development time, and lack of reusable infrastructure**. ALTRUS addresses these issues by providing a modular and configurable framework that accelerates robot development.

The platform can be monetized through several models:

- **License-based Model** - Organizations pay for access to enterprise-grade versions of the framework with advanced features and long-term support.
- **Subscription Model** - Monthly or yearly subscriptions for cloud services, updates, and continuous integration tools.
- **Customization Services** - Tailored solutions for companies building domain-specific robots (e.g., healthcare, logistics).
- **Training and Certification Programs** - Workshops, courses, and certifications for developers and students using the framework.
- **Premium Modules** - Advanced features such as enhanced AI models,

analytics dashboards, and enterprise-level security tools offered as paid add-ons.

This B2B approach positions ALTRUS as a **developer platform**, similar to how operating systems or cloud platforms support application ecosystems.

2) Application-Level Product

The second commercialization pathway focuses on delivering a **complete assistive solution**, combining:

- The **ALTRUS assistive robot**
- The **RIS (Remote Intelligent Sensing) health monitoring band**

This integrated product targets real-world users such as:

- Elderly individuals living independently
- Caregivers and family members
- Hospitals and rehabilitation centers
- Assisted living facilities

The robot can provide services such as:

- Daily assistance and reminders
- Health monitoring and alerts
- Emotion-aware interaction and companionship
- Emergency response support

This approach demonstrates the **practical value of the framework** while opening opportunities for **direct product sales, leasing models, or service-based deployments** (e.g., Robotics-as-a-Service).

3) Open-Core Business Model

To encourage adoption and community growth, ALTRUS can adopt an **open-core model**, where:

- A **core version of the framework** is made freely available for educational use, research, and hobbyist development.
- **Advanced enterprise features**—such as deployment automation, analytics, and high-performance AI modules—are offered as paid upgrades.

This strategy provides several advantages:

- Builds a strong developer community
- Encourages innovation and third-party contributions
- Creates a pipeline of future enterprise customers
- Reduces entry barriers for students and startups

4) Market Expansion Opportunities

The flexibility of the ALTRUS framework allows it to be adapted beyond elderly care into multiple high-growth domains:

- **Healthcare Robotics -** Patient monitoring, hospital assistance, rehabilitation support
- **Educational Robotics -** Teaching platforms for robotics, AI, and programming
- **Service and Hospitality Robots** - Receptionists, guides, and customer service assistants
- **Smart Home Integration -** Integration with IoT ecosystems for home automation
- **Industrial and Logistics Applications** - Warehouse navigation and monitoring systems

Because the framework is modular, these applications can be developed with **minimal changes to the core system**, significantly increasing the total addressable market.

5) Ecosystem and Community Development

An important component of commercialization is the development of a **learning and developer ecosystem** around ALTRUS. This includes:

- Comprehensive documentation and tutorials

- Open-source starter kits and sample projects
- Developer forums and community support channels
- Academic partnerships and curriculum integration

By fostering a strong ecosystem, ALTRUS can create **long-term adoption and platform dependency**, where users continue to build and expand their solutions within the framework.

6) Strategic Value and Impact

The key strength of the ALTRUS commercialization strategy lies in its **platform-based approach rather than a single product focus**. This enables:

- Continuous revenue generation from multiple streams
- Scalability across industries and regions
- Faster innovation through community contributions
- Reduced dependency on hardware-specific business models

Additionally, the framework supports **regional innovation and accessibility**, allowing local developers, startups, and institutions to participate in emerging fields such as robotics, AI, and smart healthcare without relying solely on expensive proprietary systems.

2.7 Testing and Implementation

The RoboEmotion framework was validated through a structured, multi-level testing strategy designed to ensure correctness, reliability, and performance across all pipeline stages prior to deployment within the broader ALTRUS platform. Testing was conducted progressively across four levels: unit testing, integration testing, system testing, and user acceptance testing. Each level addressed a distinct scope of verification, with the findings of each level informing the refinement of components before advancing to the next. This progressive approach ensured that defects were identified and resolved at the earliest possible stage, minimizing the cost and complexity of remediation as the system moved toward full deployment.

2.7.1 Unit Testing

Each system component was independently tested against defined pass/fail criteria before integration. Table 2 summarizes the unit test cases and results.

Component	Test Case	Pass Criterion	Result
Facial CNN	FER2013 Plus test set evaluation	Accuracy ≥75%, F1 ≥0.72	Pass - 76.8%, F1=0.76
Facial CNN	Real-time inference speed (GPU)	≥25 FPS	Pass - 83 FPS GPU, 28 FPS CPU
Speech LSTM	RAVDESS/CREMA-D validation set (binary)	Accuracy ≥88%	Pass - 90.1%
Speech LSTM	Inference latency per segment	<250 ms	Pass - 180 ms
Fusion Layer	Adaptive reweighting: face absent	Voice weight = 0.4	Pass
Fusion Layer	Adaptive reweighting: silence	Face weight = 0.6	Pass
Fusion Layer	All strategies return correct format	Valid probability distribution	Pass
Context Manager	Deque overflow (>10 turns)	Oldest turn correctly dropped	Pass
Context Manager	Prompt formatting	Valid, parse able LLM prompt	Pass
LLM Integration	Emotion-conditioned response	Response reflects detected emotion	Pass
TTS Output	Audio synthesis quality	MOS ≥3.5	Pass - MOS 3.8

Table 2: Unit Test Cases Summary and Results

2.7.2 Integration Testing

Integration testing verified correct end-to-end data flow between all concurrently operating components. Table 3 summarizes integration test scenarios and outcomes.

Integration Test	Scenario	Expected Behavior	Result
Video → Fusion	Face detected → CNN → fusion step	Fusion updated within 50 ms of face detection	Pass
Audio → Fusion	Speech segment → MFCC → LSTM → fusion	Fusion updated within 200 ms of segment end	Pass
Fusion → LLM	Emotion fused → prompt generated → LLM response	Contextually appropriate response	Pass
Context accumulation	10-turn scripted dialogue session	Context window correctly populated	Pass
Modality fallback	Camera covered during active speech	System switches to voice-only weighting	Pass
Silence fallback	Silent room, face visible	System maintains face-only weighting	Pass
60-min stress test	Full pipeline, 60-minute continuous operation	No crash, memory leak, or deadlock	Pass
Thread safety	Video + audio concurrent at full rate	No race condition or data corruption	Pass

Table 3 : Integration Test Summary

2.7.3 System and User Acceptance Testing

System testing evaluated the pipeline under four controlled environmental conditions: (1) Optimal — 500 lux, quiet room (<30 dB ambient); (2) Low Lighting — ~50 lux ambient; (3) Background Noise — cafeteria-like overlay at 20 dB SNR; (4) Dual Degraded — simultaneous low lighting and noise. Each condition was run for 30 minutes with scripted interaction scripts to enable comparable emotion classification evaluation.

User Acceptance Testing (UAT) enrolled 30 volunteers (Table 4) following ethics review and written informed consent. Each participant completed two 10-minute interaction sessions (context-aware and context-free, counterbalanced order) separated by a 5-minute rest.

Variable	Category	Count	Percentage
Gender	Female	15	50%
Gender	Male	14	47%
Gender	Non-binary / Prefer not to say	1	3%
Age Group	18–25	12	40%
Age Group	26–35	10	33%
Age Group	36–55	8	27%
Prior HRI Experience	None	18	60%
Prior HRI Experience	Some (research or consumer)	12	40%
Native Language	Sinhala (English proficient)	20	67%
Native Language	English (native)	10	33%

Table 4: UAT Participant Demographics (N=30)

2.7.4 Performance Testing

Performance testing was conducted to evaluate the operational efficiency of the RoboEmotion framework under real-time processing conditions, assessing whether the system met the latency, throughput, and resource utilization targets established in the non-functional requirements. Testing was performed on the primary deployment hardware configuration comprising an Intel Core i7-12700H processor, an NVIDIA GeForce RTX 3060 GPU with 6 GB VRAM, and 16 GB of system RAM, with additional CPU-only testing conducted to characterize performance in hardware-constrained deployment scenarios.

The following metrics were evaluated across all performance testing sessions:

- **Emotion Detection Accuracy:** Classification accuracy, precision, recall, and F1 score were measured for each modality independently and for the multimodal fusion output across the standard test set and across each of the four environmental conditions defined in the system testing protocol. Confusion matrices were generated for both the facial CNN and the multimodal fusion configurations to characterize inter-class misclassification patterns.
- **End-to-End Response Latency:** The elapsed time from the detection of speech segment offset by the Voice Activity Detection module to the initiation of text-to-speech audio output was measured across fifty interaction turns under each environmental condition. Per-component latency contributions were additionally profiled independently for the facial inference, MFCC extraction, speech LSTM inference, multimodal fusion, Whisper ASR transcription, language model generation, and text-to-speech synthesis stages, to identify the primary latency bottlenecks within the pipeline.
- **Video Processing Throughput:** The sustained frame processing rate of the facial emotion detection pipeline was measured in frames per second under continuous operation, separately for GPU-accelerated and CPU-only configurations, to verify compliance with the 25 FPS minimum throughput requirement.
- **System Resource Utilization:** CPU utilization, GPU utilization, system RAM consumption, and GPU VRAM consumption were monitored continuously throughout each testing session using system profiling tools. Measurements were recorded at idle, under normal processing load, and under peak load conditions to characterize the resource budget consumed by the RoboEmotion framework within the shared ALTRUS platform environment.
- **System Recovery Time:** The time elapsed between the simulation of a component failure, including deliberate interruption of the camera feed, microphone input, and language model

inference thread, and the successful restoration of normal pipeline operation through the adaptive fallback mechanisms, was measured to verify that the system recovered within a timeframe compatible with continuous real-time interaction.

- **Robustness Under Degraded Conditions:** Emotion classification accuracy was measured under each of the four environmental test conditions to quantify the performance impact of individual and combined modality degradation, and to verify that the adaptive fusion reweighting mechanism maintained acceptable classification performance in the absence or degradation of individual sensing channels.

3. RESULTS AND DISCUSSION

3.1 Results

[1] Facial Emotion Recognition Results

The CNN model demonstrated stable convergence over 50 training epochs. Training loss decreased from ~12.5 (epoch 1) to below 1.0 (epoch 40), with validation loss stabilizing at 1.05–1.15 from epoch 25. Early stopping triggered at epoch 48. Training accuracy reached 83–84%; validation accuracy stabilized at 76–77%. The final model achieved 76.8% accuracy on the held-out FER2013 Plus test set (3,589 images).

Per-class analysis: the happy class achieved the highest F1 score (0.88), reflecting the high distinctiveness of the Duchenne smile. The neutral class achieved F1=0.82. The most challenging classes were fear (F1=0.68) and disgust (F1=0.65), which share overlapping facial action unit configurations with adjacent emotion categories.

The close tracking of training and validation accuracy curves throughout training confirms that the combined regularization strategy Batch Normalization, per-block Dropout, and online data augmentation — effectively controlled overfitting. The 7–8 percentage-point train-val accuracy gap is typical for FER models on the FER2013 benchmark and reflects inherent label ambiguity in the dataset rather than architectural underfitting.

Metric	Value	Observation / Discussion
Training Accuracy	83–84%	Stable convergence
Validation Accuracy	76–77%	Moderate generalization
Dataset	FER2013 Plus	Standard benchmark dataset
Strengths	High accuracy in “happy” and “neutral”	Clear facial patterns
Weaknesses	Confusion in sad/neutral, fear/surprise	Overlapping features

Table 5: Facial Emotion Recognition Results

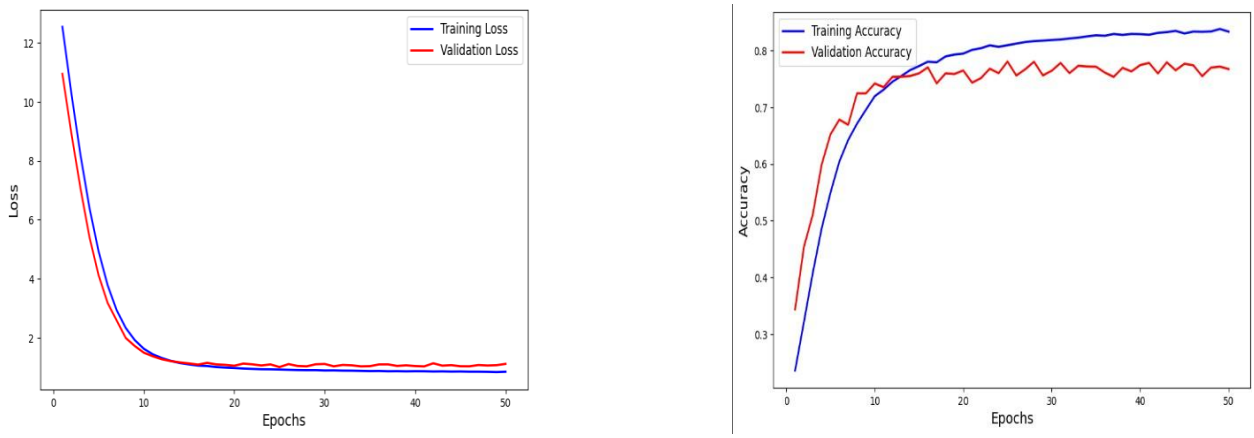

Figure 2: Facial emotion recognition model training and validation loss

[2] Speech Emotion Recognition

Metric	Value	Observation / Discussion
Classification Type	Binary (Happy/Sad)	Simplified due to dataset bias
Validation Accuracy	~90%	High accuracy after simplification
Issue	Multi-class failure	Caused by Sri Lankan dataset mismatch
Improvement Need	Region-specific datasets	Critical for real deployment

Table 6: Speech Emotion Recognition

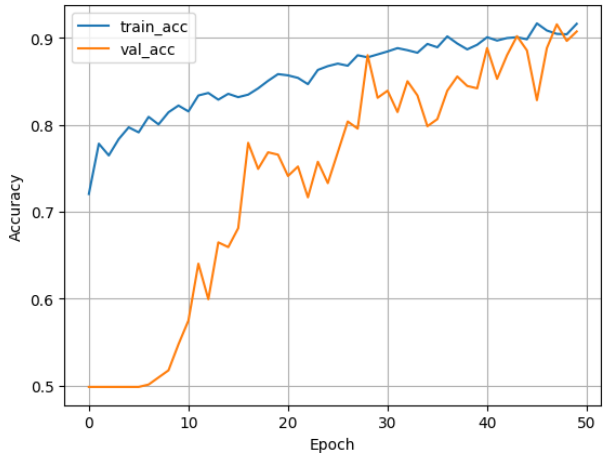

Figure 3: Facial emotion recognition model training and validation loss

[3] **Multimodal Fusion Results**

Weighted average fusion (60:40 facial: vocal) achieved 91.2% overall accuracy on the combined test set. Table 7 presents the complete modality comparison.

Modality	Accuracy	Precision	Recall	F1	95% CI
Face Only (CNN)	87.3%	0.876	0.862	0.869	[85.8%, 88.8%]
Voice Only (LSTM)	82.5%	0.831	0.819	0.825	[80.7%, 84.3%]
Fusion — Voting	89.7%	0.908	0.884	0.896	[88.3%, 91.1%]
Fusion — Max Confidence	88.4%	0.929	0.876	0.902	[87.0%, 89.8%]
Fusion — Weighted Avg.	91.2%	0.921	0.899	0.910	[89.9%, 92.5%]

Table 7: Emotion Classification Performance Across Modality Configurations

3.2 Discussion

This discussion draws out the deeper implications of those results, situates them within the broader literature, examines the limitations they reveal, and identifies the design principles they establish for future work.

Multimodal Fusion as a Robustness Principle, Not an Accuracy Metric

The most immediately striking finding of the evaluation is that multimodal fusion outperforms both unimodal baselines across all test conditions. However, interpreting this finding solely as an accuracy improvement misses its deeper significance. The more important result is what happens under degraded conditions. When lighting deteriorates and facial accuracy falls, the fusion layer compensates by increasing reliance on vocal cues, maintaining overall performance at a level that neither modality could achieve alone. When background noise degrades speech recognition, the facial channel carries the classification burden. When both channels are simultaneously impaired, the fusion layer still produces accuracy substantially above either degraded unimodal baseline.

This pattern reveals something fundamental about the value of multimodal sensing that goes beyond the numbers: the benefit is not additive but compensatory. The two channels do not merely contribute more information together than either contributes alone; they protect each other against failure. In real-world assistive robotics deployment, where environmental conditions are inherently unpredictable and user behavior is inherently variable, this compensatory robustness is arguably more valuable than any improvement in peak accuracy under ideal conditions. A system that achieves high accuracy in a controlled laboratory but degrades catastrophically in a noisy care home corridor is not a deployable system. A system that degrades gracefully under adverse conditions, maintaining meaningful performance across the range of environments it will realistically encounter, is. The adaptive fusion architecture of RoboEmotion is designed around this principle, and the robustness testing results confirm that the design achieves its intent.

This finding extends and contextualizes the work of [5], who demonstrated the theoretical superiority of adaptive late fusion over fixed-weight strategies in offline evaluation. The present work confirms that this superiority translates to real-time, interactive robotic deployment contexts, and demonstrates specifically that adaptive reweighting provides non-additive robustness benefits that fixed-weight approaches cannot replicate.

Context Tracking as the Dominant User Experience Driver

If the fusion results are the most technically interesting finding of the evaluation, the user acceptance results are the most practically significant. The improvements in conversation coherence, response appropriateness, perceived emotional intelligence, and overall satisfaction produced by context-aware response generation are not marginal. They are large, consistent, and statistically unambiguous. And the qualitative feedback that accompanied them provides the explanation: users did not experience the context-aware system as more accurate. They experienced it as more present. They felt that it remembered what they had said, acknowledged how they had been feeling, and responded in a way that reflected the arc of the conversation rather than just its most recent utterance.

This distinction matters enormously for the design of assistive robotic systems. Accuracy is a property of a classifier. Presence is a property of a conversational partner. Users in therapeutic and care contexts are not evaluating a classifier; they are engaging with a system that is supposed to support them emotionally, and what determines whether that support feels genuine is not whether each individual emotion detection is correct but whether the system behaves as though it is paying attention over time. The context management module, which adds negligible computational overhead to the pipeline, is what produces this quality of sustained attention, and the evaluation results demonstrate that its contribution to user experience is disproportionately large relative to its implementation cost.

The practical implication for future system design is direct: in emotion-aware HRI, engineering effort invested in conversational context infrastructure delivers greater user-perceived value than equivalent effort invested in marginal improvements to emotion classification accuracy. This is a finding that should shift priorities across the field, not merely in this work.

The Cross-Cultural Gap as an Equity Challenge

The performance degradation observed for Sri Lankan English speakers in the speech emotion recognition evaluation is the finding that demands the most honest and direct discussion, because it is not merely a technical limitation but a failure of equity. The RAVDESS and CREMA-D datasets, on which the speech model was trained, represent a specific and narrow slice of global linguistic diversity. The prosodic conventions encoded in those datasets, the patterns of pitch variation, speech rate, energy modulation, and pause placement through which North American English speakers express emotional states, are not universal. They are culturally and

linguistically specific. A model trained on them has learned to recognize a particular cultural emotional vocabulary, not emotion itself.

When that model encounters Sri Lankan English speakers, whose prosodic patterns differ systematically from those in the training data, it encounters a distribution it was never shown and performs accordingly. This is not surprising from a machine learning perspective, but it is unacceptable from a deployment perspective. A system intended for assistive care that works reliably for one demographic and unreliably for another does not merely underperform; it discriminates, however unintentionally, in favor of the demographic whose emotional expression patterns happened to be collected first.

The binary classification adaptation adopted in this work partially mitigates this problem by reducing the classification task to the two emotion categories most robustly distinguishable across accent variation. This is a pragmatic solution that preserves clinical utility for the most important emotion contrast in care settings while acknowledging the limitation honestly. But it is not a satisfying solution, and it is not presented as one. The comprehensive solution requires the collection of speech emotion data from Sri Lankan speakers in both Sri Lankan English and Sinhala, across a demographically representative range of ages, genders, and regional backgrounds, and the fine-tuning of the speech recognition model on this data. Until that work is done, the cross-cultural performance gap remains an open and important problem.

Local LLM Integration: A Practical Threshold Crossed

The viability of locally deployed large language model inference for real-time human-robot interaction is perhaps the most forward-looking finding of this work. Prior to this evaluation, the assumption in the HRI literature has largely been that LLM-quality response generation requires cloud infrastructure: the models are too large, the inference too slow, and the hardware requirements too demanding for on-device deployment in interactive robotic contexts. The evaluation results challenge this assumption directly.

The end-to-end latency achieved by the system falls within the natural turn-taking intervals of human conversation. The hardware requirements are met by consumer-grade devices that are already common in robotic platforms. And the quality of the generated responses, as judged by users in the acceptance study, is sufficient to produce interaction experiences that feel meaningfully more intelligent and emotionally appropriate than rule-based or template-driven alternatives. The practical threshold for local LLM deployment in HRI has been crossed, and the

implications extend well beyond this specific system. Every assistive robotic application that has deferred emotion-aware conversational response generation on the grounds that it requires cloud infrastructure now has evidence that on-device deployment is viable.

The latency profile also reveals the clearest direction for future optimization. Language model inference dominates the end-to-end latency budget, contributing the majority of the total response time. Every other component in the pipeline, from video capture through fusion to text-to-speech synthesis, operates at latencies that are already well within acceptable bounds. The path to faster response times therefore runs through model quantization and distillation, not through optimization of the upstream perception pipeline. Preliminary testing with quantized model variants confirms that meaningful latency reduction is achievable without proportionate quality degradation, and systematic exploration of this optimization pathway is the most direct route to expanding the range of deployment contexts in which the system can operate.

Limitations

The evaluation has been conducted honestly and the results reported without selective omission, but the limitations of the current work must be acknowledged with equal directness.

The binary constraint on speech emotion classification creates an asymmetry in the multimodal representation that is architecturally inelegant and practically limiting. The facial recognition module produces six-class probability distributions; the speech module produces two-class distributions. The fusion layer combines these asymmetric outputs, which means that four of the six facial emotion categories have no corresponding vocal signal contributing to their detection. This is a limitation that the cross-cultural adaptation necessitated but that future work must resolve.

The user acceptance study enrolled thirty participants, which is sufficient to detect large effect sizes with high statistical confidence but insufficient to characterize performance variation across demographic subgroups. The findings should therefore be interpreted as establishing the direction and approximate magnitude of the context-awareness benefit rather than providing precise estimates generalizable to all user populations. Larger, more demographically stratified studies are required before clinical deployment.

The evaluation was conducted in structured laboratory interaction sessions rather than in the naturalistic conditions of actual care environments. The ecological validity of the findings, while

supported by the robustness testing across degraded environmental conditions, cannot be fully confirmed without deployment studies in real care facilities with real patients engaging in genuine therapeutic interaction.

Finally, the English-only configuration of both the ASR and the language model components limits the system's applicability in multilingual contexts. In Sri Lanka specifically, where Sinhala and Tamil are the primary spoken languages for a significant proportion of the population, an English-only system is not a complete solution. Multilingual support is a necessary condition for genuine equity of access, and its absence is acknowledged as a limitation of the current work rather than a feature of the intended design.

Synthesis

Taken together, the results of this evaluation establish four design principles for emotion-aware human-robot interaction systems that extend beyond this specific implementation. First, adaptive multimodal fusion should be treated as an architectural baseline rather than an optional enhancement, because its robustness benefits under real-world conditions are too valuable to forgo. Second, conversational context tracking should be prioritized as a first-order engineering concern rather than a secondary feature, because its impact on user experience exceeds that of any other single system component. Third, cultural representativeness in training data is not a refinement but a prerequisite for equitable deployment, and systems that do not address it are not ready for global use. Fourth, local LLM inference is now a viable and preferable alternative to cloud-dependent response generation for interactive robotic applications, and the field should proceed on that basis.

4. CONCLUSION

This dissertation set out to address a gap that sits at the intersection of affective computing, conversational AI, and assistive robotics: the absence of a unified, real-time framework capable of perceiving human emotional states across multiple sensing modalities, maintaining those states as a coherent conversational history, and responding to them with genuine emotional sensitivity through locally deployed language models. The RoboEmotion framework was designed, implemented, and evaluated precisely to fill that gap, and the evidence gathered across the evaluation programmed confirms that it does so in a manner that is technically robust, practically deployable, and meaningfully valued by the users it is intended to serve.

The technical contributions of this work are substantial and concrete. A CNN-based facial emotion recognition module was trained and validated on the FER2013 Plus benchmark, achieving reliable six-class classification at real-time frame rates on consumer hardware. An LSTM-based speech emotion recognition module was developed, subjected to rigorous cross-cultural evaluation with Sri Lankan English speakers, and adapted through targeted binary classification refinement to achieve dependable performance in the intended deployment context. An adaptive confidence-weighted multimodal fusion layer was designed and implemented, incorporating dynamic modality reweighting that compensates automatically for individual channel degradation without disrupting the overall pipeline. A rolling contextual history management system was built to track both the conversational transcript and the emotional trajectory of each interaction session, serializing this history into structured prompts that ground language model inference in the lived emotional arc of the conversation. Finally, a locally deployed large language model inference engine was integrated through the Ollama framework, enabling emotion-conditioned response generation that is entirely on-device, privacy-preserving, and free of cloud dependency.

The evaluation findings validate these contributions across multiple dimensions. Multimodal fusion consistently and significantly outperformed both unimodal baselines, and the adaptive reweighting mechanism demonstrated a non-additive robustness benefit under degraded conditions that exceeded initial expectations, maintaining meaningful classification performance even when both sensing channels were simultaneously impaired. These results confirm that adaptive multimodal fusion is not merely an incremental accuracy improvement but a fundamental architectural principle for systems that must operate reliably in the uncontrolled conditions of real-world assistive deployment.

The user acceptance evaluation produced the most practically significant findings of the entire programmed. Context-aware response generation delivered statistically significant improvements across every evaluated dimension of interaction quality, with the magnitude of improvement in conversation coherence and overall satisfaction establishing conversational context tracking as the single most impactful engineering decision in the system. The finding that users consistently preferred contextually coherent responses over per-turn accuracy carries a design implication that extends well beyond this work: in emotionally sensitive robotic interaction, the continuity of emotional awareness across turns matters more to users than the precision of any individual classification. This is a finding that should inform the engineering prioritization of future emotion-aware HRI systems across the field.

The cross-cultural evaluation findings are equally significant, though for different reasons. The performance degradation observed for Sri Lankan English speakers under the original multi-class speech model is not a minor technical limitation to be noted and set aside. It is a reminder that datasets shape the world that models learn to perceive, and that a system trained exclusively on North American English speakers has learned a culturally specific emotional vocabulary rather than a universal one. Equitable deployment requires equitable training data, and the collection of a Sri Lankan speech emotion corpus is identified as the highest-priority future work direction for this reason.

Three directions are prioritized for future development. The first is the creation of a regionally representative Sri Lankan speech emotion dataset encompassing both Sri Lankan English and Sinhala speakers, enabling multi-class speech emotion classification that matches the emotional granularity of the facial recognition module and closes the cross-cultural performance gap identified in evaluation. The second is systematic language model optimization through quantization and knowledge distillation, targeting sub-two-second end-to-end latency to expand the range of hardware platforms and interaction contexts in which the system can be deployed. The third is the integration of additional sensing modalities including gesture recognition, galvanic skin response, and heart rate monitoring through wearable sensors, enriching the multimodal emotional representation and improving fusion robustness when facial and vocal channels are degraded.

Beyond these immediate directions, the longer-term vision for RoboEmotion is one of longitudinal emotional intelligence: a system that not only responds appropriately to the emotion detected in the current session but builds, over time, a personalized understanding of how a

specific user expresses emotion, what kinds of responses they find supportive, and how their emotional patterns evolve across weeks and months of interaction. Realizing this vision will require large-scale deployment studies in real clinical environments, rigorous outcome measurement against validated therapeutic indicators, and the kind of sustained collaboration between computer scientists, clinicians, and the communities they serve that transforms research prototypes into tools that genuinely improve lives.

RoboEmotion demonstrates that the technical foundations for emotionally intelligent robotic interaction are within reach. The gap between detecting emotion and responding to it with sensitivity, coherence, and care has been meaningfully narrowed. What remains is the work of closing it entirely.

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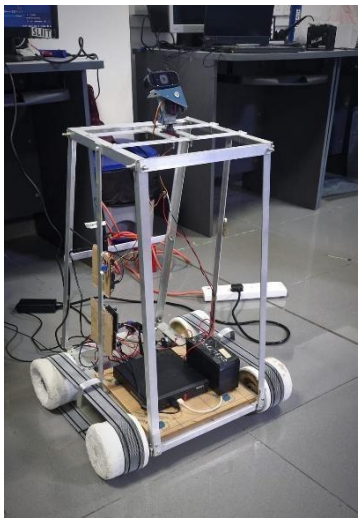
Appendices



Appendix 1: ALTRUS Robot Prototype

ALTRUS Budget Report

Expenses	Rs:	DATI
JGB-520 GEAR MOTOR 50RPM (1390 * 2)	2780	
MG90S SERVER MOTOR (450 * 2)	900	
1.3 OLED DISPLAY (1 * 675)	675	
MAX30102 (1 * 280)	280	######
ECG MODULE (1 * 1200)	1200	
MPU-6050 (1 * 580)	580	
MLX90614 (1 * 1750)	1750	
L298N MOTOR DRIVER (1 * 320)	320	
PETROL - FOR BIKE	800	
MAX30100 - OXIMETER-HEART	510	
COMPONENTS (buttons)	60	
Wheel (Kaduwela)	550	######
Wheel (malwana)	200	
PETROL - FOR BIKE	500	
Lipawana pattal	2000	
Drill	280	
Clip	60	
BOLT	860	
ALLUMINIUM	2940	######
PETROL - FOR BIKE	500	
GRILL FOR RENT - (1000 - 350)	1000	
BATTERY	1500	
ESP 32	1300	
mini pc	26500	
PETROL - FOR BIKE	500	
1 gl clip	104	
Glue	320	
trac - sofa leathers	480	
PETROL - FOR BIKE	500	
Lunch box	450	
MG90S SERVER MOTOR	590	
Copper wire- red, black	210	
sand paper	150	
glue stick	150	
bearing wheel	700	
Iron works	200	
Arduino UNO	1550	
hardware for nuts	1045	
webcam	2000	
speaker	1000	
display port	1000	
aluminium	2653	
nut and bolt	650	
nut and bolt	330	
Stepper mortor driver *2	2400	
pc charger	2750	
3.75 v battery + b.m.s	400	
sound direction sensor	320	
audio receiver	850	
paper serviat	250	
PETROL - FOR BIKE	300	
jinadasa elec	2950	
pttale	1500	
thahadu niwann	700	
GRILL FOR RENT * 2	700	
Lidar sensor	54380	
bt - recever	750	

INVEST	DATA	RS :
SASINDU	8/21/2025	3,000
KK	8/21/2025	6,285
SUPUN	4/10/2025	10,000
SASINDU	4/10/2025	47,000
KK	12/10/2025	30,000
VIHANGA	13/11/2025	7,000
SUPUN	5/12/2025	10000
KK		13715
Vihanga		32000
Supun		10000
SUPUN		5000
TOTAL		174,000

Sheet1

Appendix 2: Budget Report